



Low-Cost Water Level Monitoring for Municipal Water Management

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Commercial Devices

- Upfront Costs and Maintenance Cost
- Data Transmission Limitations
- High Power Requirements
- Proprietary Software



Project Background

*Proceedings of IEEE International Conference on Applied System Innovation 2018
IEEE ICASI 2018- Meen, Prior & Lam (Eds)*

A Continuous Water-Level Sensor Based on Load Cell and Floating Pipe

Sheng-Wei Wang; Chen-Chia Chen*; Chieh-Ming Wu; Chun-Ming Huang

National Chip Implementation Center
7F, No. 26, Prosperity Rd. I, Science Park
Hsinchu City, Taiwan

Accomplishments

High Resolution Depth
Measurements

Remote Data Transmission

Limitations

Cost Prohibitive

Not Portable

High Power Requirements

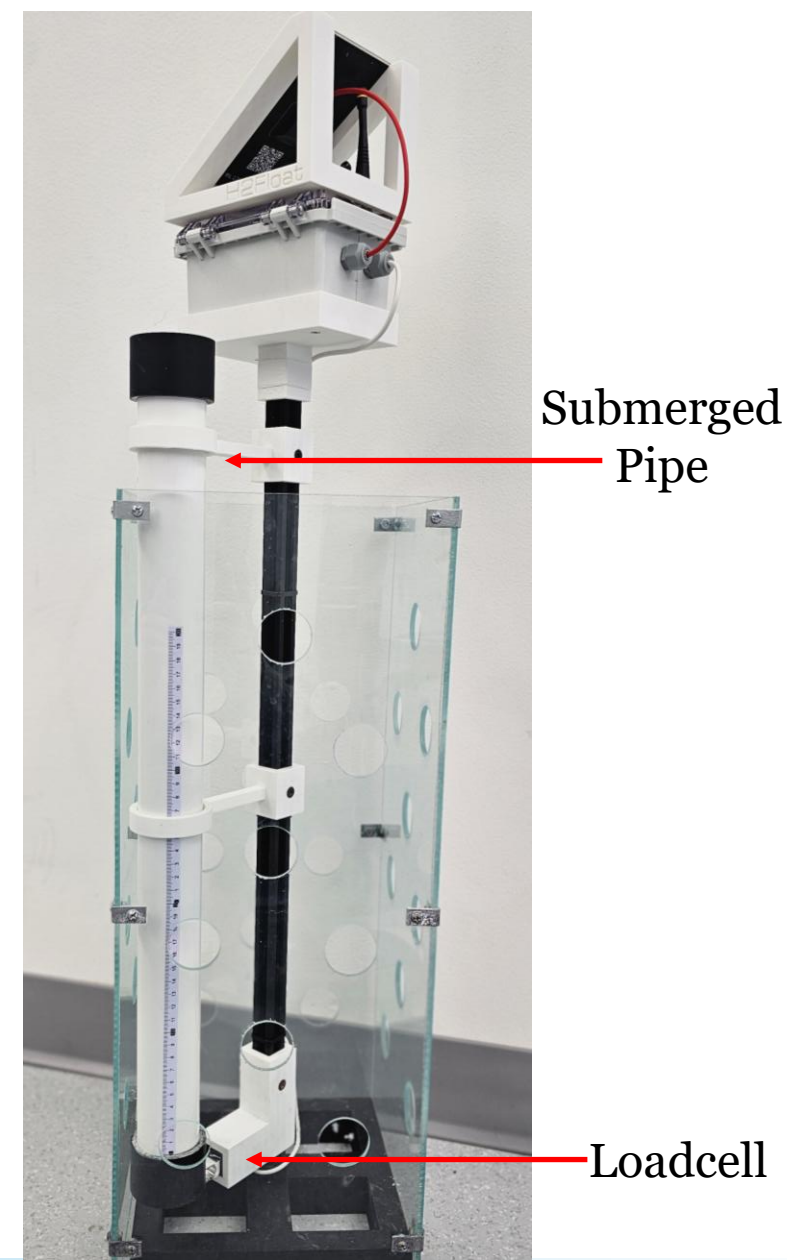


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Improved Water Level Monitoring System H₂Float

- Remotely Monitor Water Levels
- High Resolution Measurements
- Low Power Consumption
- Low Cost
- Site Specific Adaptability
- Increased Location Monitoring



Design Basis... Archimedes Principle

$$F_B = \rho g(d\pi r^2) \quad d = F_B / \rho g\pi r^2$$

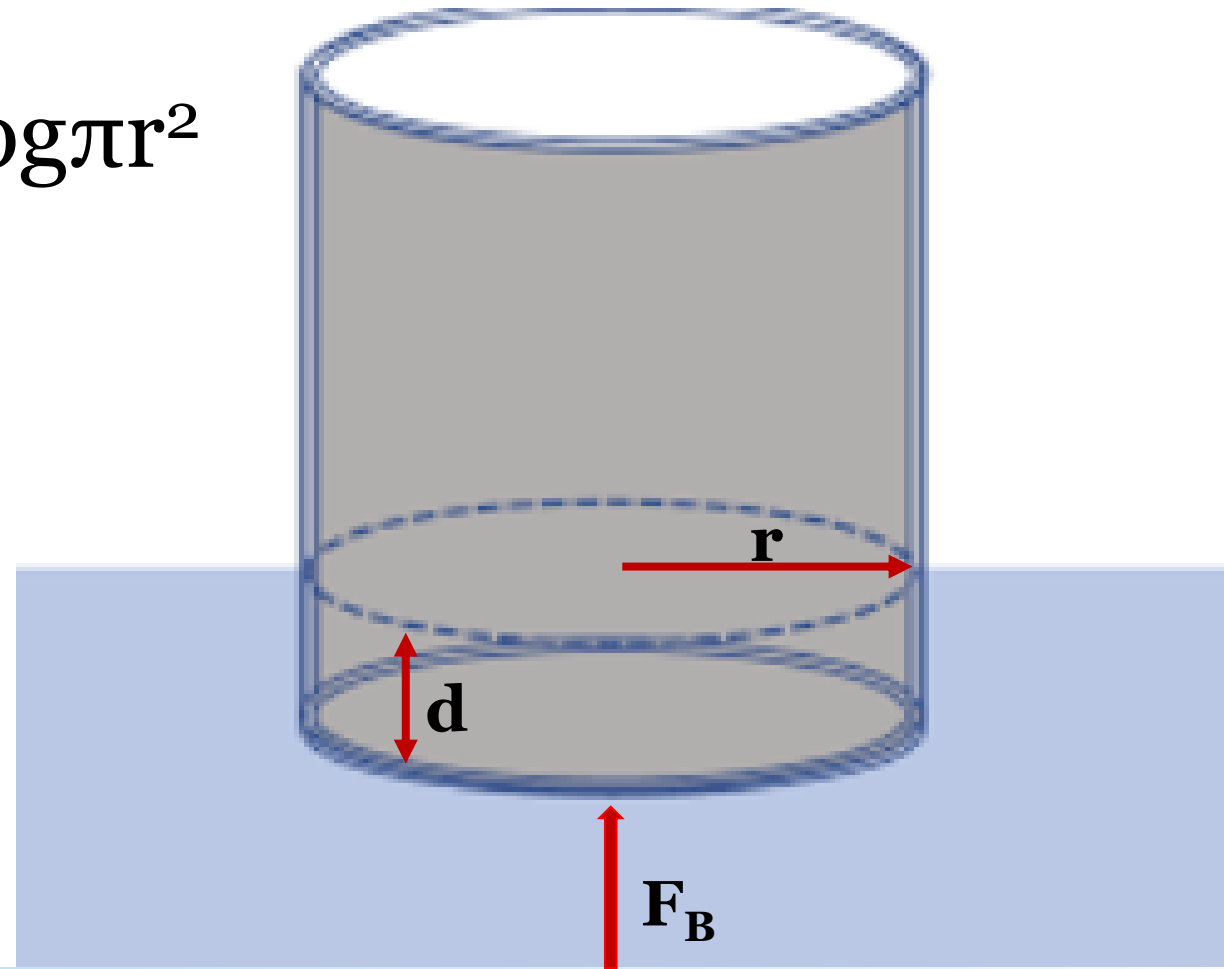
F_B = Buoyant Force

ρ = Water Density

g = Gravitational Acceleration

r = Pipe Radius

d = Submerged Depth



Calibrating H₂Float

2-point Calibration

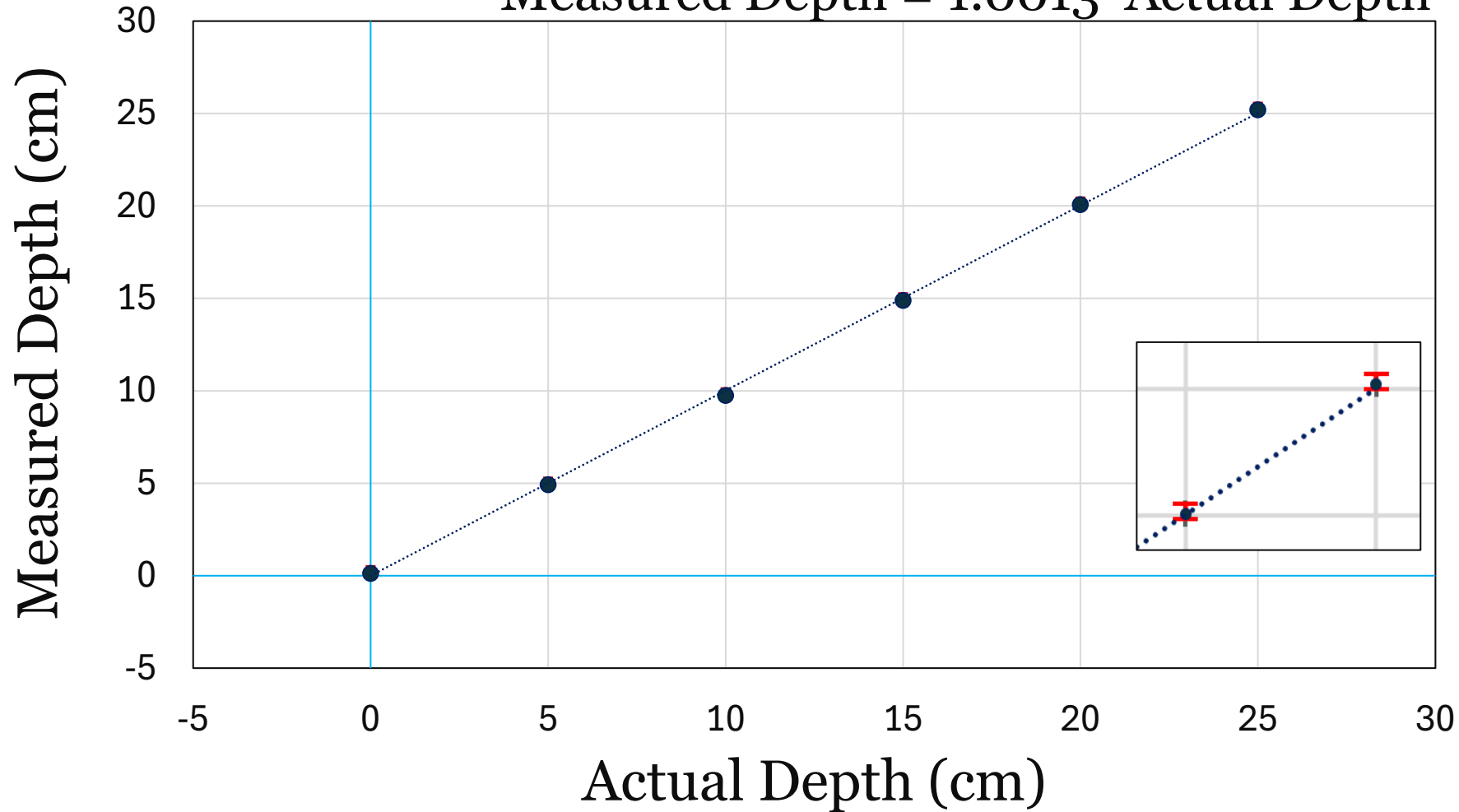
- Reading at zero weight (Tare)
- Reading at 750g

Device is calibrated once during deployment setup



Depth Testing

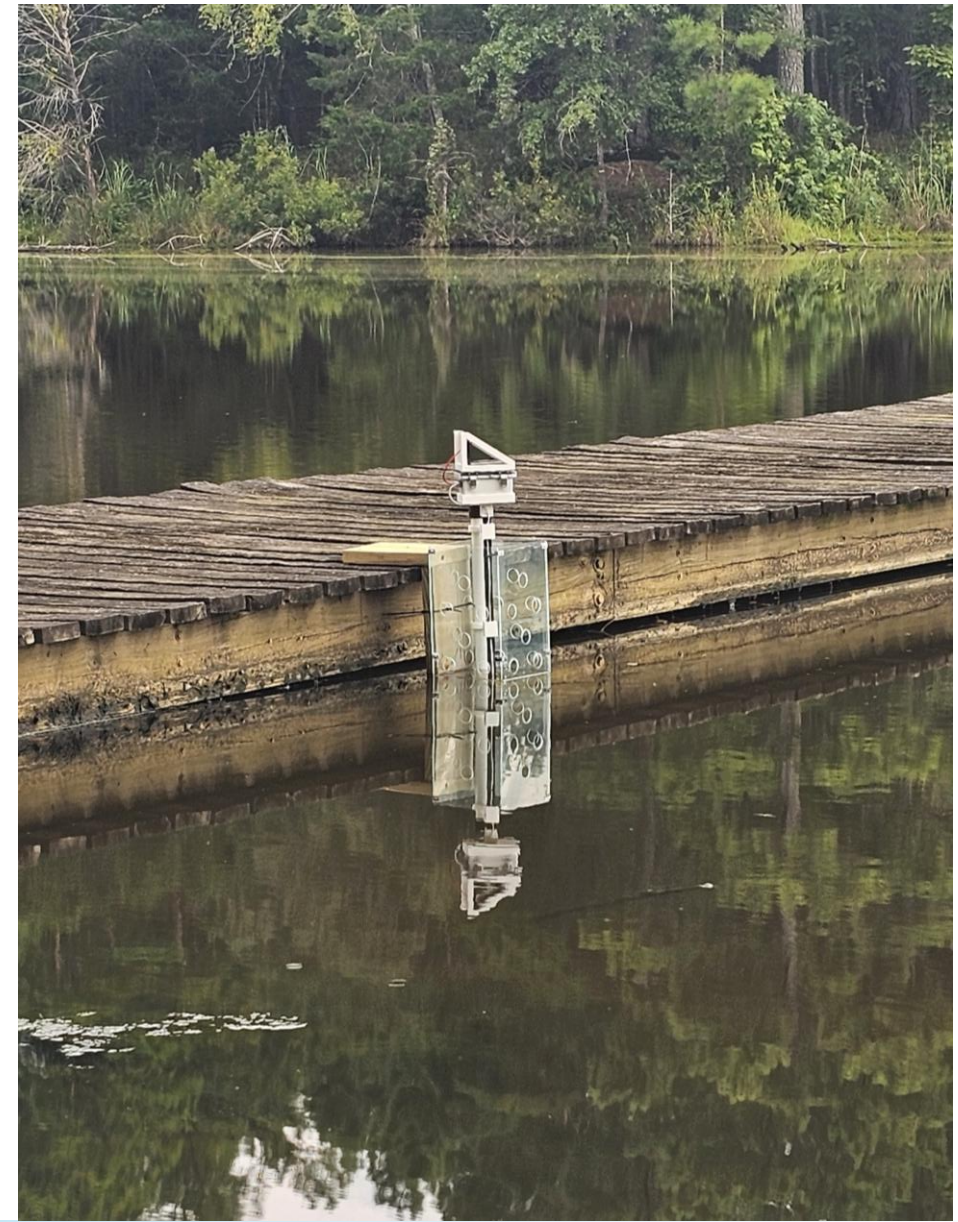
$$\text{Measured Depth} = 1.0013 * \text{Actual Depth}$$



H₂Float Prototype Testing

Initial Deployment Achievements

- ✓ Remote Data Transmission
- ✓ Autonomous Operation
- ✓ High Resolution Measurements
- ✓ Low Power
(Sleep and Wake)
- ✓ Solar Powered
- ✓ Maintained Battery Voltages
- ✓ Recorded Backup Data on SD Card



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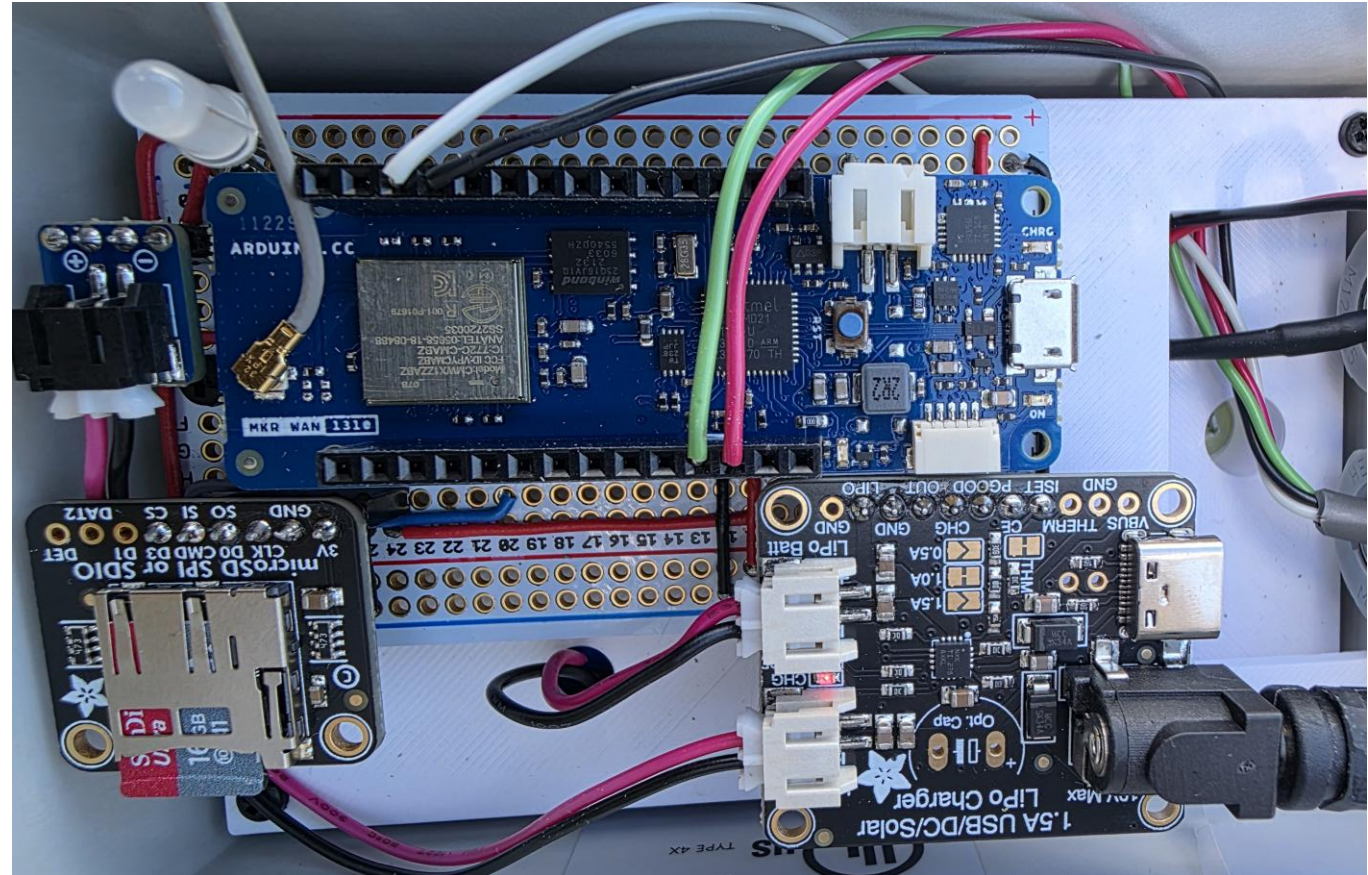
Production Model

- Self-Contained Compact Design
- Lithium Polymer Battery with Solar Cell Integration
- Low Power Consumption
- Wireless Communication
- SD Card Data Backup

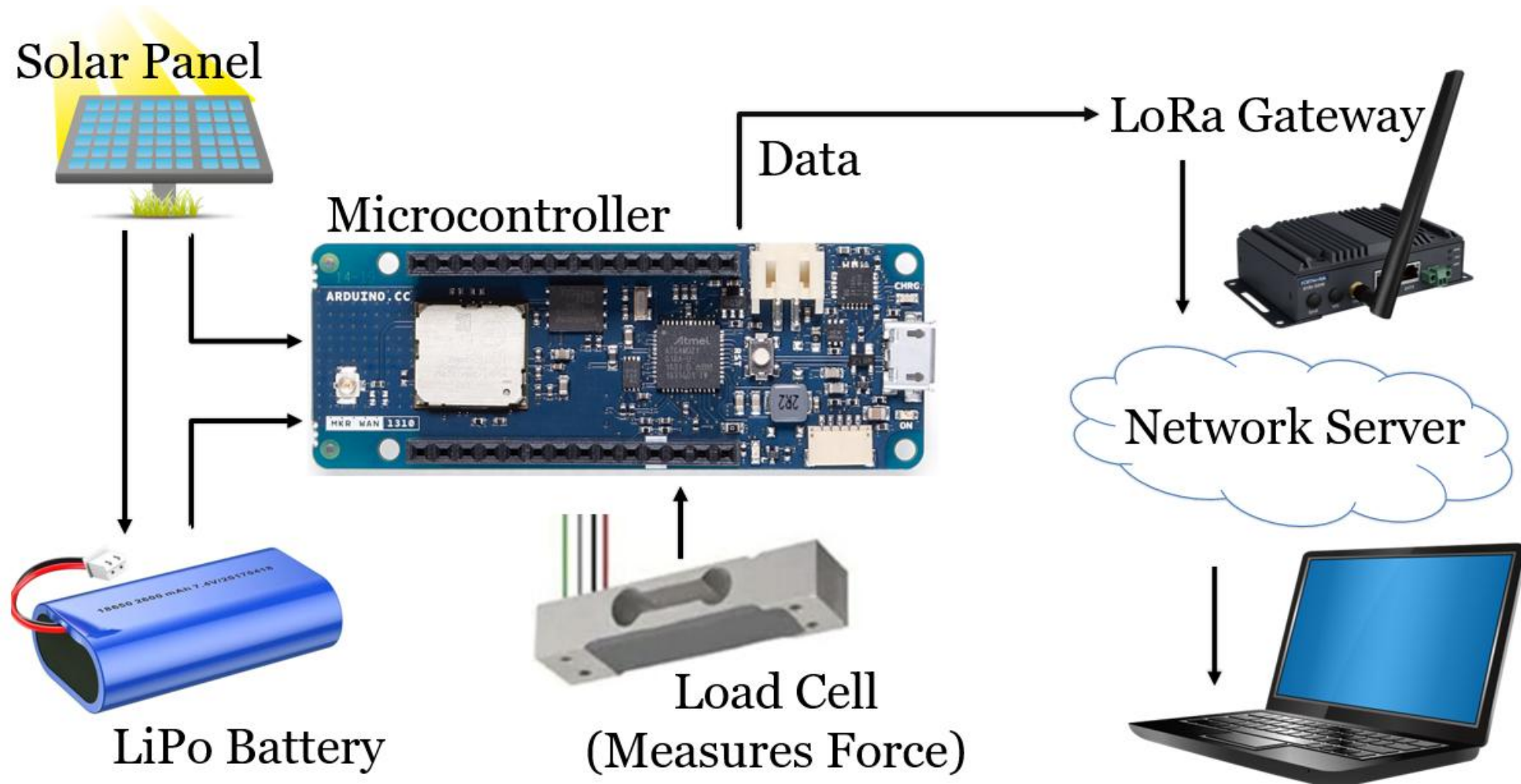


Arduino Hardware and LoRa

- Transmits Data via Long Range (LoRa) Radio
- Operates in Remote Areas
- Requires Manual Firmware Updates
- Best for Easily Accessed Locations

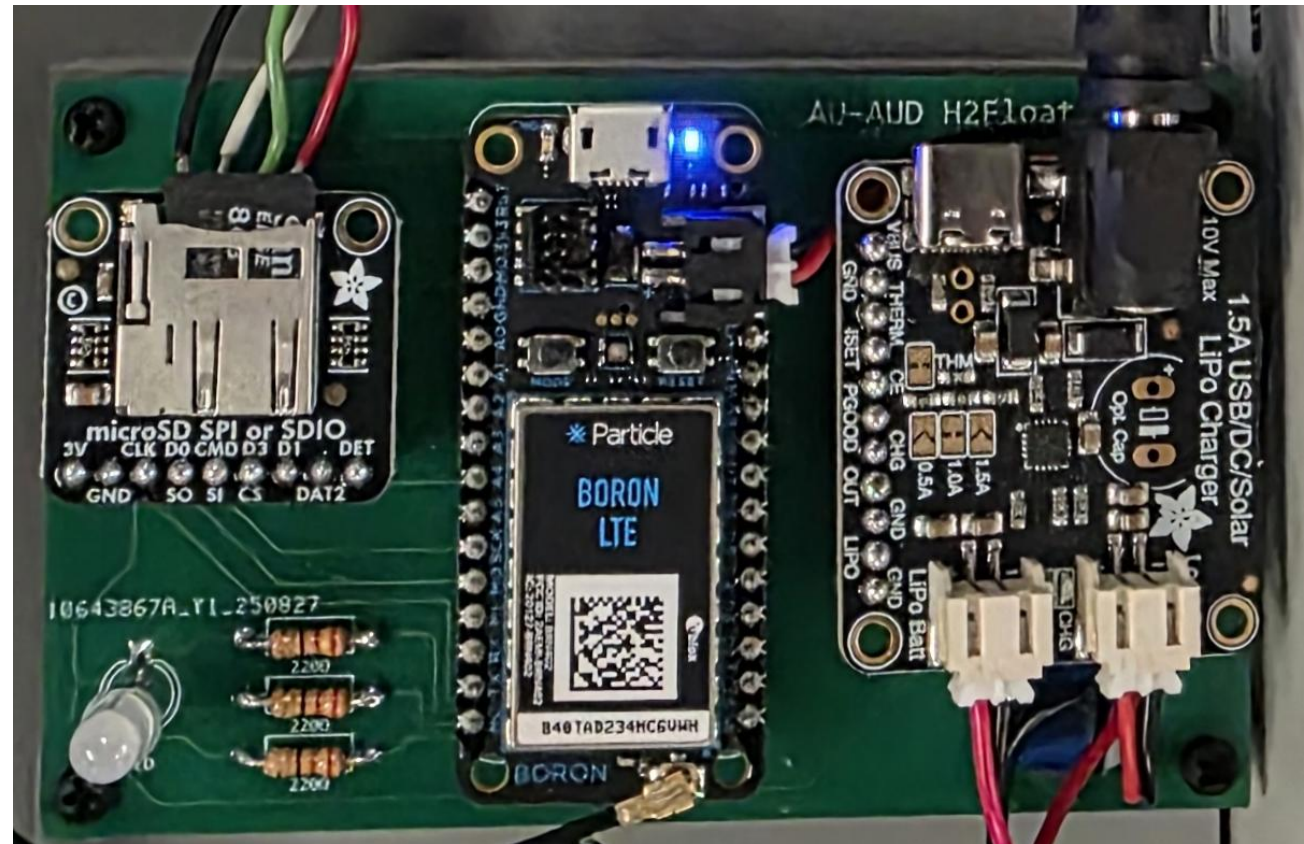


H2Float Arduino Sensor Design

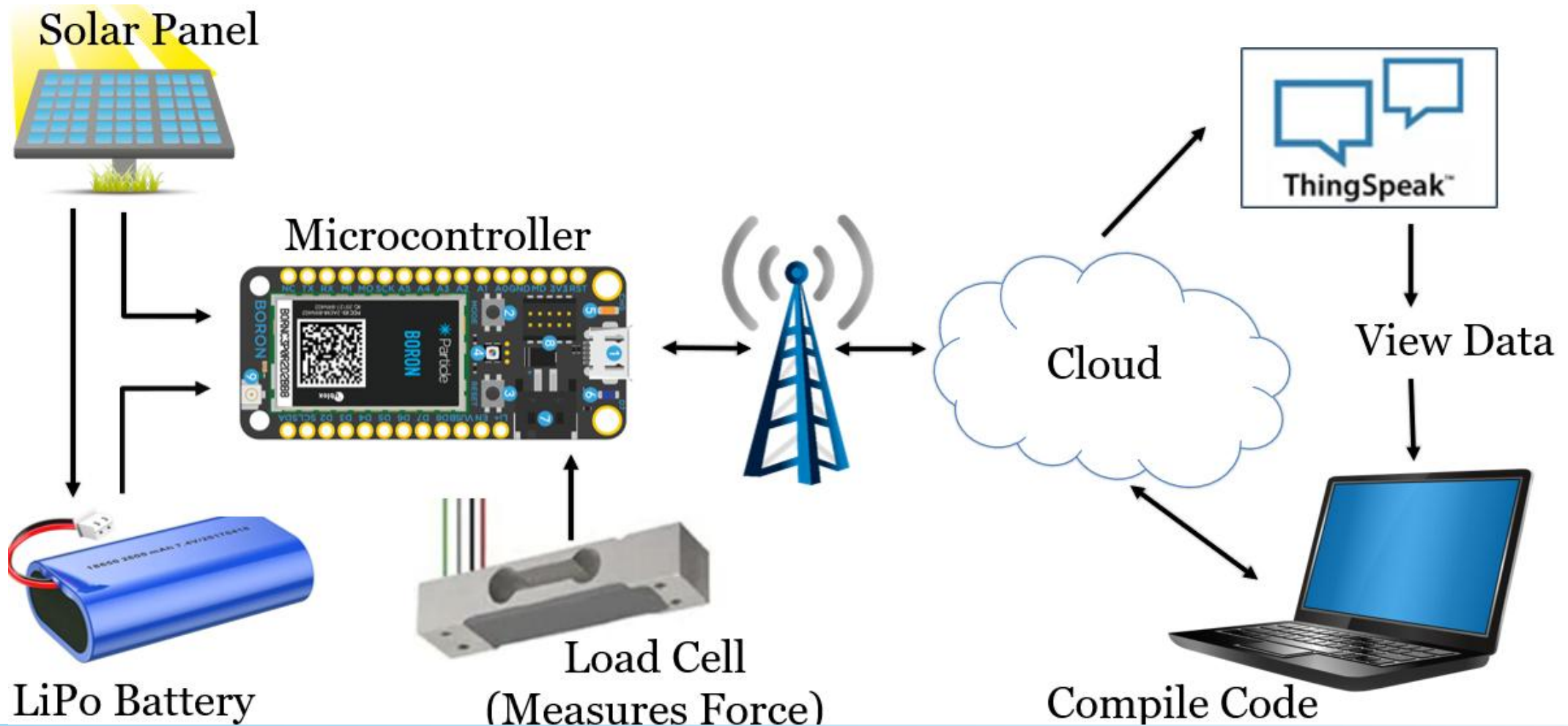


Particle Hardware and Cellular

- Cellular Network Connection
- Cellular Service Requires Monthly Fee
- Wireless Firmware Updates (OTA)
- Best for Difficult to Access Locations
- Developed PCB



H2Float Particle Boron Sensor Design

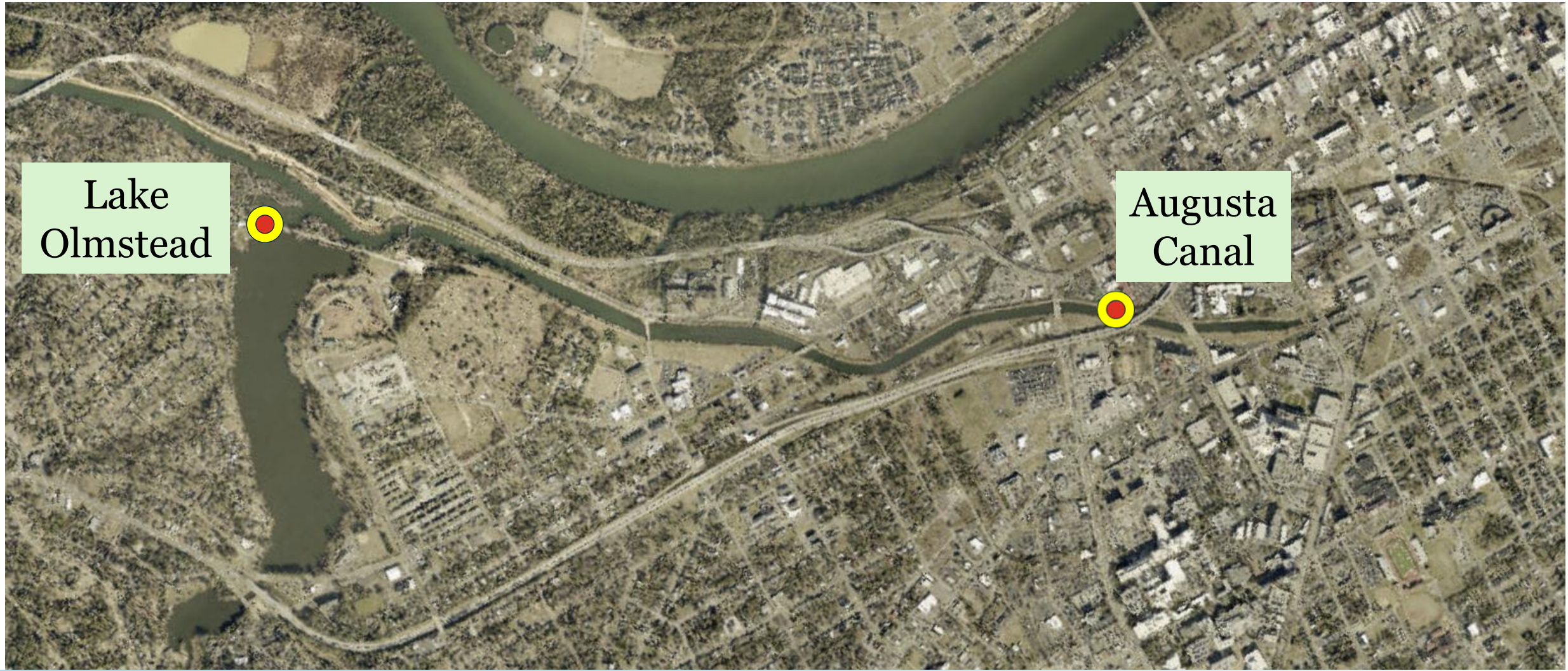


SD Card Backup Data Logging

Date	Time	Depth (cm)	Depth (in)	Depth (ft)	Sea Level (ft)	Voltage
10/1/2025	20:38:00	-0.0062	-0.0025	-0.0002	152.000	3.98
10/1/2025	20:53:04	-0.0177	-0.0070	-0.0006	151.999	3.98
10/1/2025	21:08:07	0.0048	0.0019	0.0002	152.000	3.98
10/1/2025	21:23:11	0.0251	0.0099	0.0008	152.001	3.98
10/1/2025	21:38:14	0.0874	0.0344	0.0029	152.003	3.98
10/1/2025	21:53:17	0.0942	0.0371	0.0031	152.003	3.98
10/1/2025	22:08:21	0.1043	0.0411	0.0034	152.003	3.98
10/1/2025	22:23:24	0.1297	0.0510	0.0043	152.004	3.98
10/1/2025	22:38:28	0.1892	0.0745	0.0062	152.006	3.98
10/1/2025	22:53:31	0.2089	0.0822	0.0069	152.007	3.98



Current Deployment Locations



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Augusta Canal Deployment

- 4 ft -Arduino Device
- Successful Long-Term Deployment
- Not Susceptible to Debris Encroachment

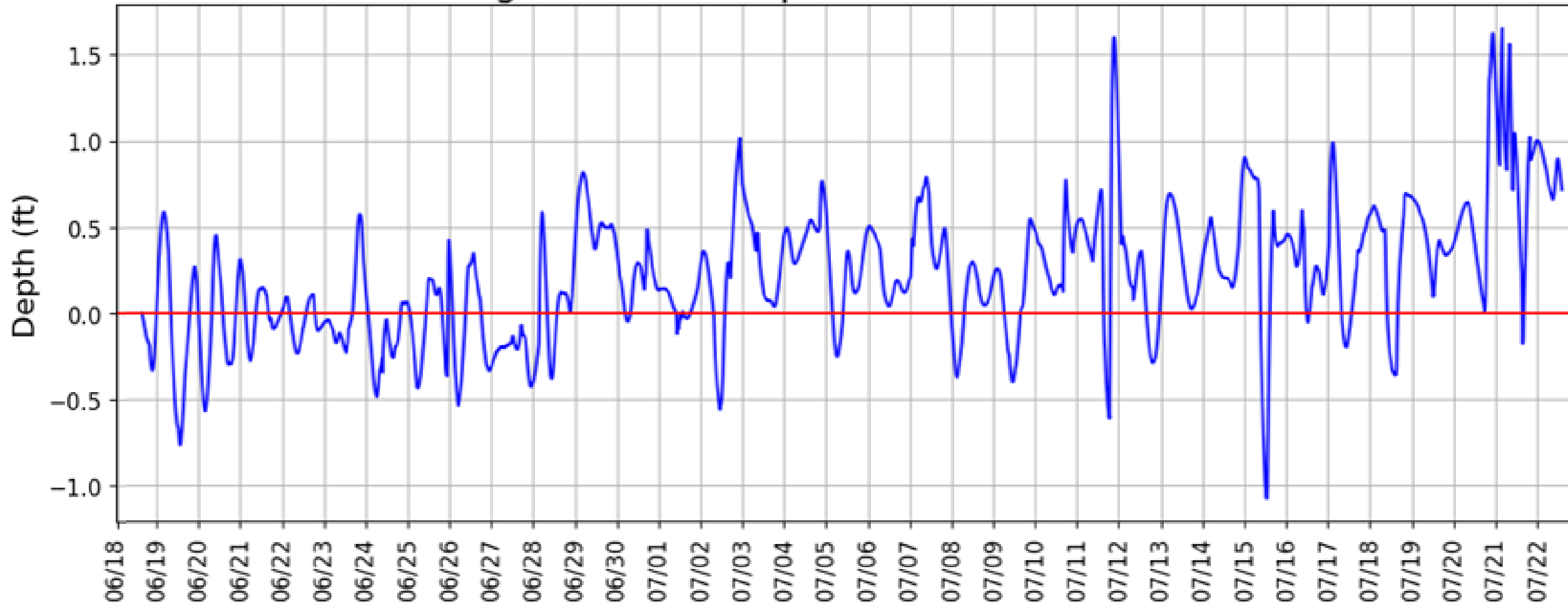


Augusta Canal Authority is using this device to maintain FERC compliance



Graphic Data

Augusta Canal Depth Variation 06/18-07/22

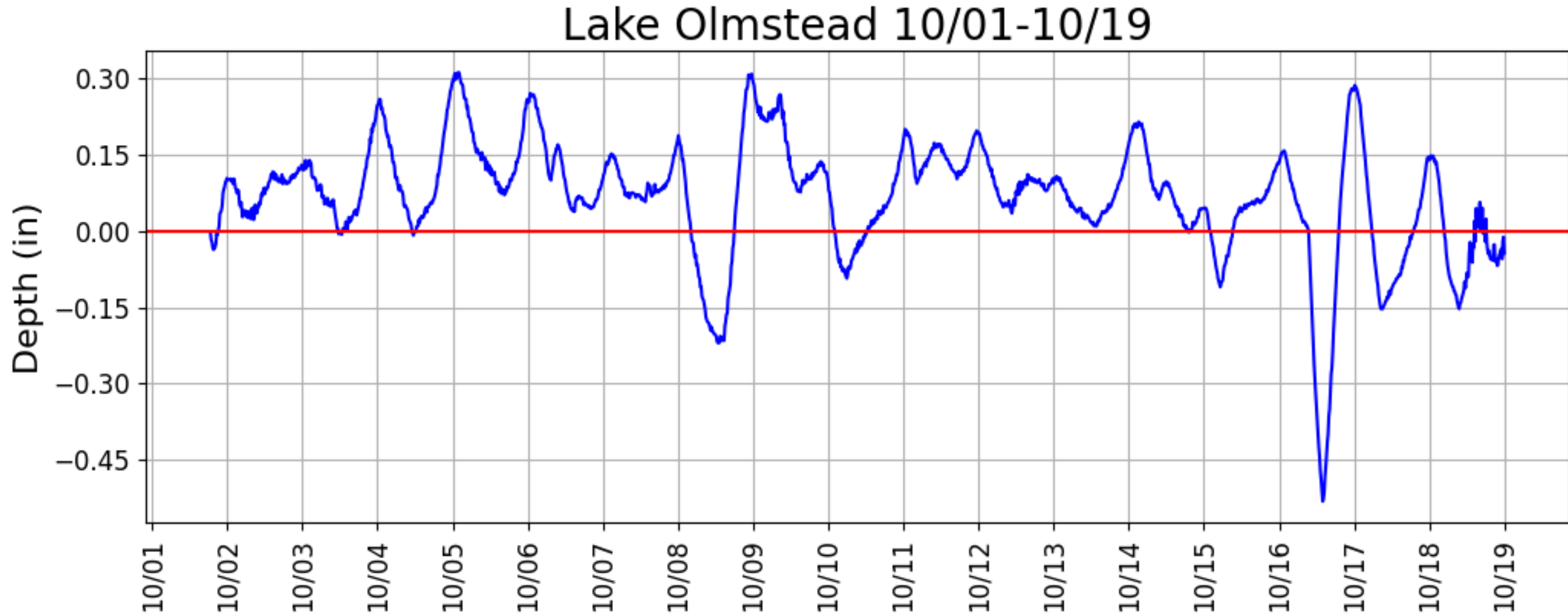


Lake Olmstead Deployment

- 6 ft - Particle Device
- Location Requires OTA Updates
- Maintains Battery Level



Graphic Data



H₂Float –Materials Cost

Microcontroller	\$46
Electronics Components	\$20
Solar	\$28
Battery	\$30
Hardware	\$62
Enclosure (waterproofing)	\$28
Calibration tool	\$13
Total Cost	\$227



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- Dr. Joseph Hauger, Augusta University
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- Dr. Oscar Flite, Augusta Utilities Engineering
- Augusta Utilities Department
- Augusta Canal Authority
- Augusta University Department of Physics and Biophysics
- AU² Team
- My husband and son



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Stronger Together

Our research group demonstrates the value of **partnership** between a public utility and a public research university.

Individually, each organization has unique capabilities and knowledge. **Together**, we're able to leverage expertise in chemistry, biology, physics, electronics, and engineering to solve challenging cross-domain problems.

Our **collaboration** so far includes autonomous waterway navigation; monitoring water chemistry, water levels, water velocity, dam seepage, and rainfall levels; and smart city solutions for trash and grass level notifications.



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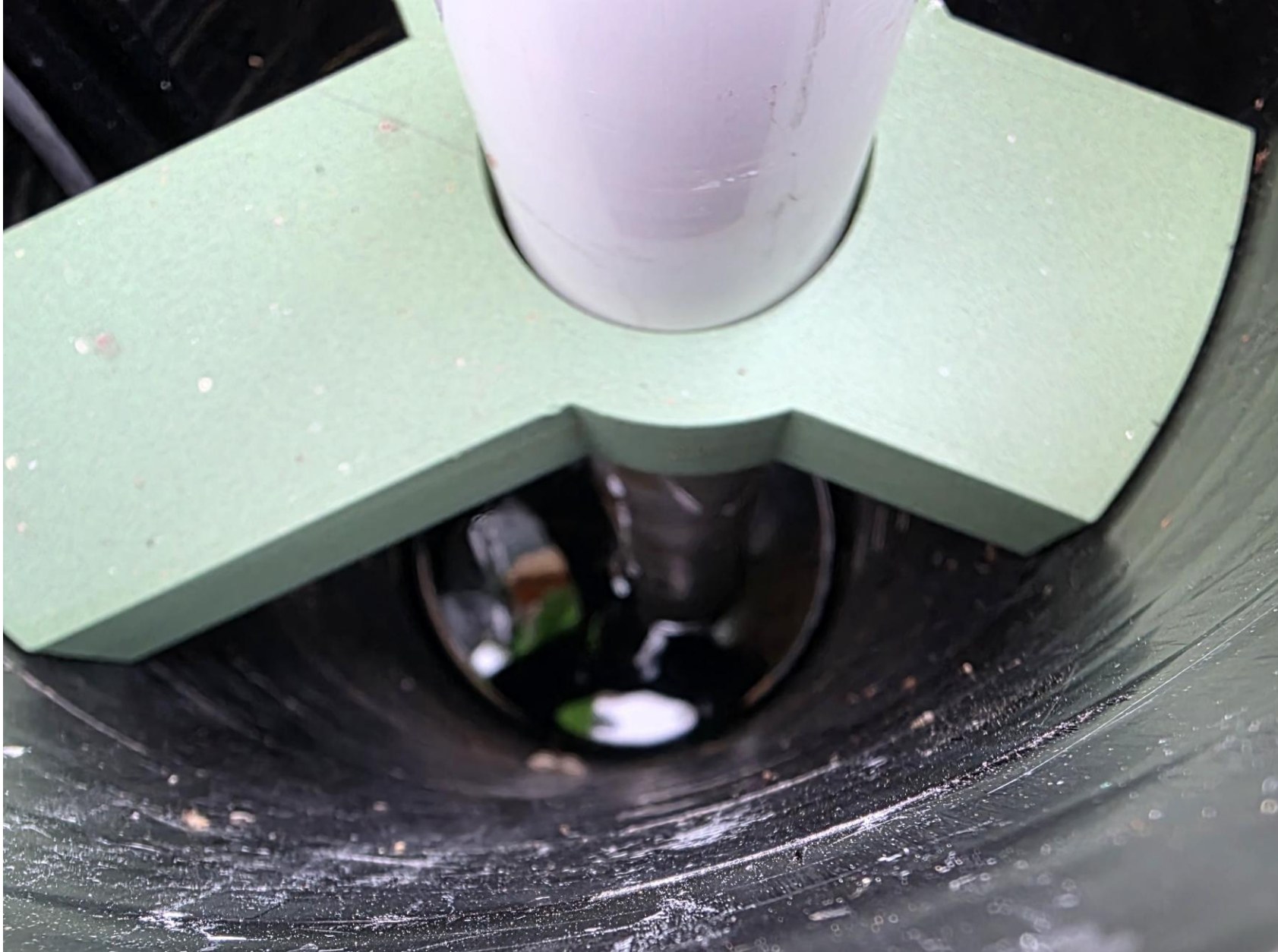
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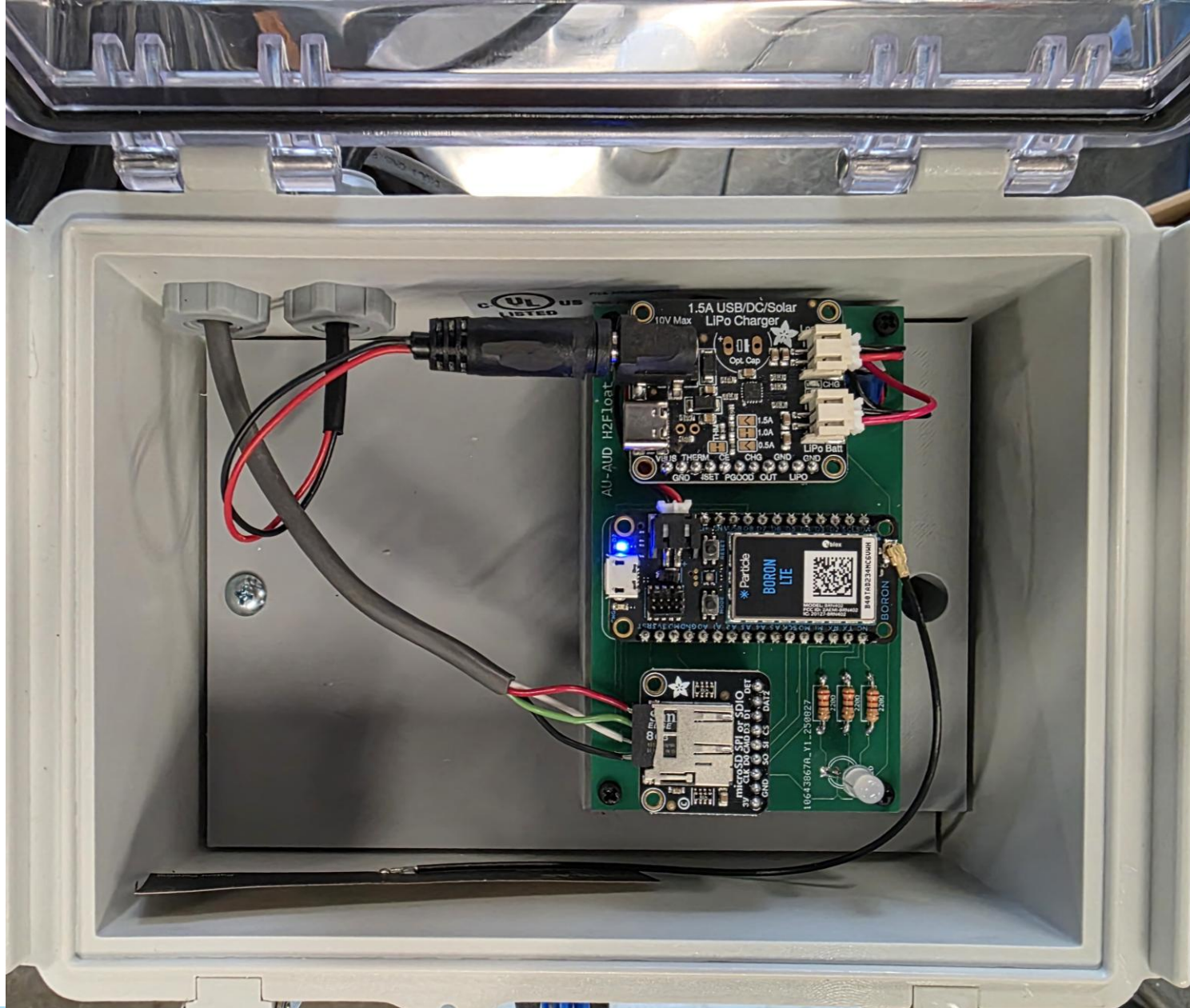
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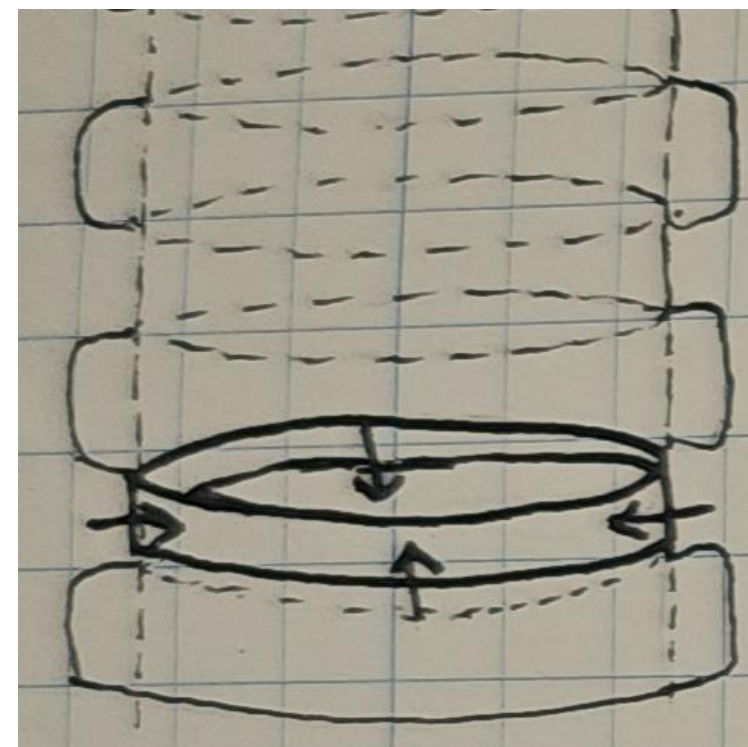
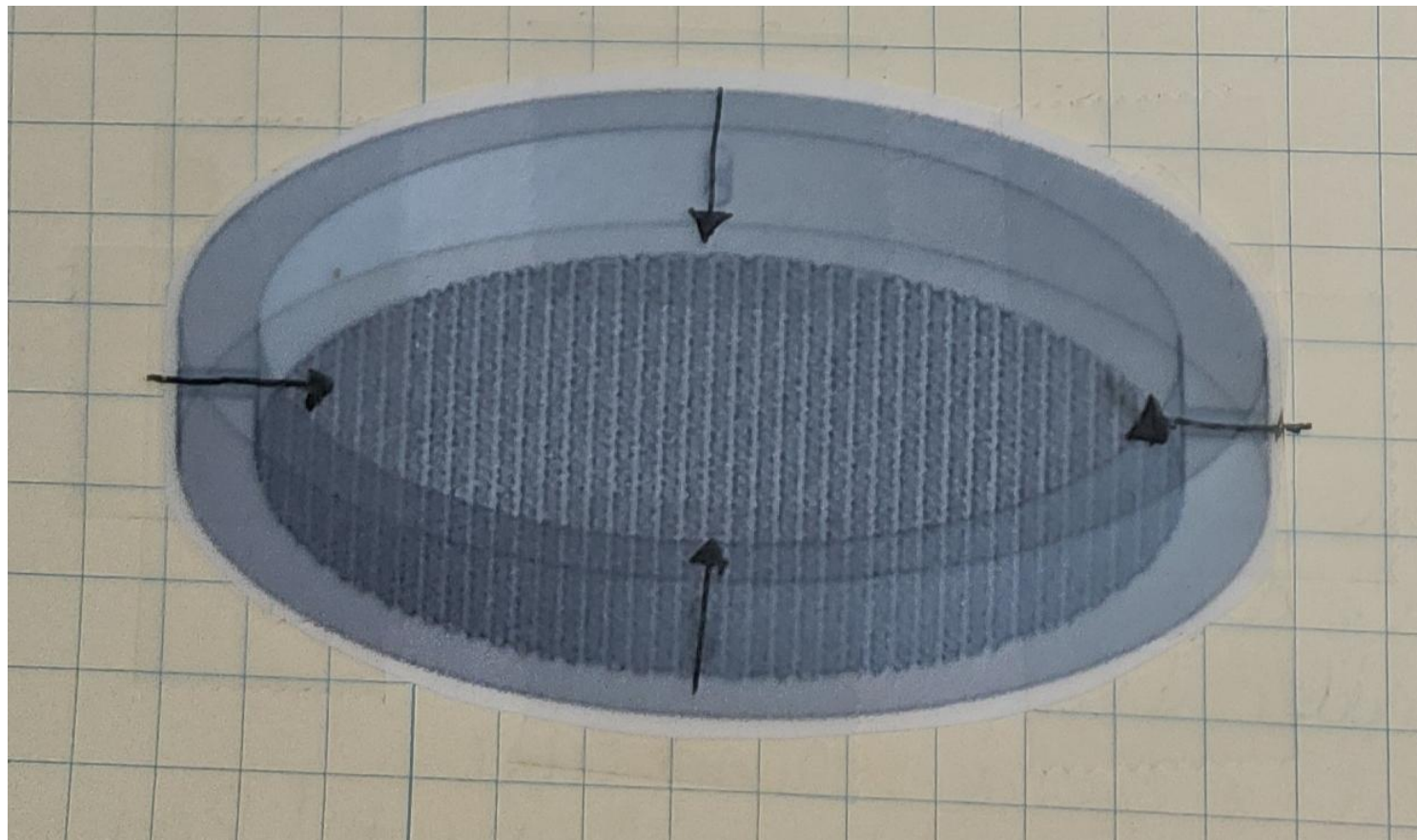


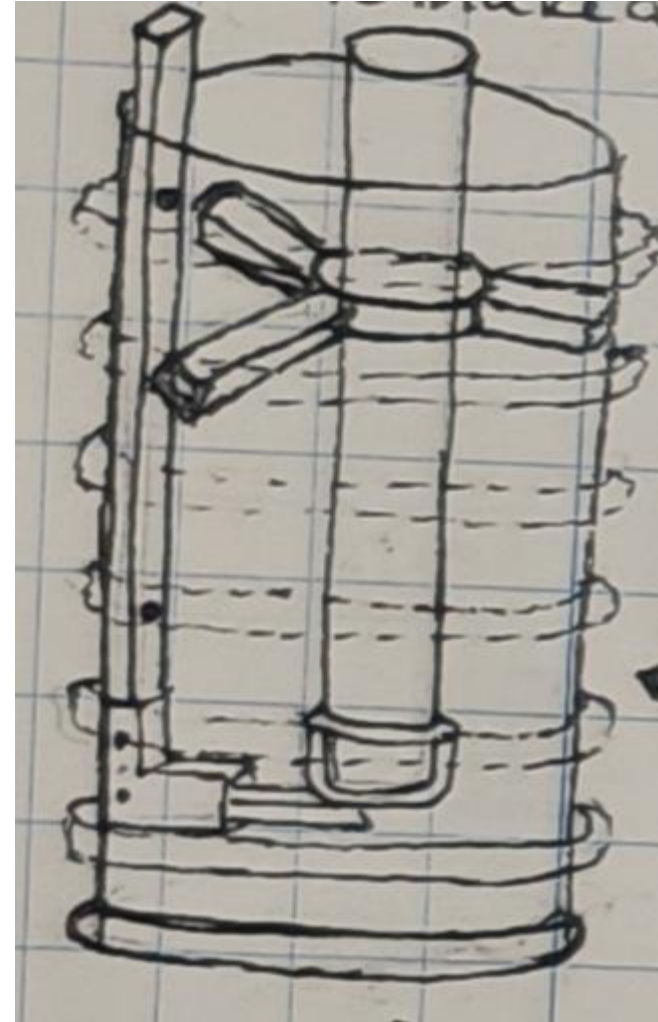
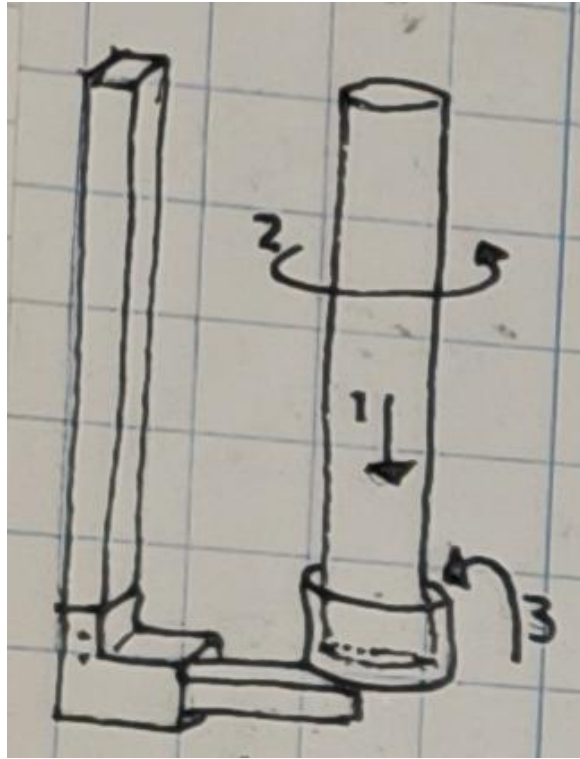


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